
Protocol Title: A Phase 1/2, Open-label, Dose-Escalation and Dose-Expansion Cohort Study of SNDX-5613 in Patients with Relapsed/Refractory Leukemias, Including Those Harboring an MLL/KMT2A Gene Rearrangement or Nucleophosmin 1 (NPM1) Mutation

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SIGNATURE PAGE

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REVISION HISTORY

Version/Date	Section	Changes implemented
Final 1.0/16Oct2020	Not Applicable	Not Applicable
Final 2.0/24Sep2021	Section 2, 3	Removed repetitive information from the protocol and referenced the protocol sections
	Section 2 (previous 4)	Updated study endpoints (removed CR after 4 weeks, replaced RFS by EFS, and added food effect endpoint)
	Section 3 (previous 5)	Clarified the primary analysis of Phase 2 part will be based on evaluable population
		Updated the Simon's 2-stage design for the adult evaluable population using the null hypothesis of 10% and target CR rate of 25% and power of 90%
		Updated the statistical design for the adult and paediatric combined (evaluable) population using null hypothesis of 10% and target CR rate of 25%
	Section 4 (previous 6)	Clarified the use of ITT, evaluable and safety populations
	Section 5.1 (previous 7.1)	Updated the derivation for EFS and DOR
	Section 6.6.2 (previous 8.6.2)	Added sensitivity analysis to provide unconditional uniformly minimum variance unbiased estimator for CR rate
	Section 6.6.3 (previous 8.6.3)	Updated TTR analysis using descriptive statistics
	Section 8.1.1 (previous 6.1.1)	Updated the interim analysis based on the design update and added the information about toxicity monitoring
	Section 9 (previous 11)	Removed repetitive information from the protocol
Final 3.0/21Feb2023	Section 2, 3	Updated based on the most recent protocol amendment
	Section 4	Updated the definition of efficacy evaluable population based on FDA feedback and added KMT2Ar Population, NPM1c Population, Formulation Effect Population, and Food Effect Population based on updated objectives and endpoints; Added a table for application of analysis population for clarity

	Section 5.1.2	Added more details on the definition of the derived efficacy endpoints
	Section 5.2.2	Added more details on how to impute missing AE end date and missing prior/subsequent anti-leukemic therapy dates
	Section 6	Clarified analysis will be performed by arm/dose in Phase 1 and by cohort in Phase 2 where applicable
	Section 6.4.4	Added a listing for the concomitant medication known to cause prolonged QTc
	Section 6.6	Added more details on the analyses based on updated hierarchical testing procedure
	Section 6.6.2	Added sensitivity analyses to consider the window between bone marrow testing and CBC testing and to assess the impact of hematopoietic growth factors and platelet transfusions, and other supportive analyses
	Section 6.6.3.3	Added sensitivity analysis for DOR corresponding to the added sensitivity analysis for CR+CRh rate
	Section 6.6.3.6	Updated the details on the analyses for transfusion independence
	Section 6.6.3.7	Updated subgroup variables
	Section 6.7.2	Added more details on the analyses for AE for patients that undergo HSCT without leaving the study and plan to resume SNDX-5613 treatment after HSCT; added more details on analyses for AE and AESI
	Section 6.7.3.2	Added summary of liver laboratory parameters
	Section 6.7.6	Updated details on QTcF categories
	Section 6.11	Added more details on formulation effect analyses
	Section 6.12 (previously Section 6.11)	Updated based on protocol amendment
	Section 6.13	Added COVID-19 related analysis
	Section 7	Updated based on the changes in the SAP

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LIST OF ABBREVIATIONS

Abbreviation or special term	Explanation
AE	Adverse Event
AESI	Adverse Event of Special Interest
ALT	Alanine aminotransferase
AST	Aspartate aminotransferase
AUC ₀₋₂₄	Area under the plasma concentration curve from time 0 to 24 hours
AUC _{0-t}	Area under the concentration versus time curve from time 0 to t
BMI	Body Mass Index
CI	Confidence Interval
C _{L/F}	Apparent oral clearance
CR	Complete remission
C _{max}	Maximum observed concentration
CRc	Composite definition of complete remission
CRh	Complete remission with partial hematologic recovery
CRi	Complete remission with incomplete hematologic recovery
CRO	Contract Research Organization
CRp	Complete remission with incomplete platelet recovery
CSR	Clinical Study Report
DBL	Database Lock
DLT	Dose-limiting toxicity
DOR	Duration of response
ECG	Electrocardiogram
ECOG	Eastern Cooperative Oncology Group
eCRF	Electronic Case Report Form
EoT	End of treatment
GCP	Good clinical practice
GMP	Good manufacturing practice
GMR	Geometric mean ratios
HIV	Human immunodeficiency virus

IC	Informed consent
ICF	Informed consent form
ICH	International Council for Harmonisation
KMTA2r	Lysine (K) Methyltransferase 2A
MedDRA	Medical Dictionary for Regulatory Activities
MLFS	Morphological leukemia-free state
MLLr	Mixed Lineage Leukemia-rearranged
MRD	Minimal residual disease
MTD	Maximum tolerated dose
MUGA	Multi-gated acquisition
NCI	National Cancer Institute
NPM1	Nucleophosmin 1
ORR	Overall response rate
OS	Overall survival
PBMC	Peripheral blood mononuclear cell
PD	Progressive disease
PFS	Progression free survival
PK	Pharmacokinetic
PO	Oral(ly)
PR	Partial remission
PT	Preferred term
R/R	Relapsed/refractory
RBC	Red blood cell count
RFS	Relapse-free survival
RP2D	Recommended Phase 2 dose
SAE	Serious Adverse Event
SAP	Statistical Analysis Plan
SD	Standard Deviation
SI	International System of Units
SOC	System Organ Class

SOP	Standard Operating Procedure
SRC	Safety Review Committee
SUSAR	Suspected unexpected serious adverse reaction
$t_{1/2}$	Half-life
TBI	Total body irradiation
TEAE	Treatment-emergent adverse events
$T_{\max}$	Time to maximum plasma concentration
TRAE	Treatment-related treatment-emergent adverse event
TTR	Time to response
TFLs	Tables, Figures and Listings
ULN	Upper limit of normal
$V_{z/F}$	Apparent volume of distribution
WBC	White blood cell count

1 INTRODUCTION

The purpose of this Statistical Analysis Plan (SAP) is to provide detailed descriptions of the statistical methods, data derivations and data displays for study protocol SNDX-5613-0700 “A Phase 1/2, Open-label, Dose-Escalation and Dose-Expansion Cohort Study of SNDX-5613 in Patients with Relapsed/Refractory Leukemias, Including Those Harboring an MLL/KMT2A Gene Rearrangement (KMT2Ar) or Nucleophosmin 1 (NPM1) Mutation” for Clinical Study Report (CSR) analysis.

Any deviations from this SAP will be described and justified in the Clinical Study Report (CSR).

The preparation of this SAP has been based on International Council for Harmonisation (ICH) E9 guidelines.

All data analyses and generation of Tables, Figures and Listings (TFLs) will be performed using SAS version 9.4® or higher.

Study objectives and design can be found in US Protocol Amendment 14.1 Version 15.1 Section 3 and 4.

2 STUDY ENDPOINTS

In Phase 1, the study endpoints will be assessed by dose cohort and overall for each arm unless otherwise stated. For Phase 2 endpoints will be assessed by disease cohort (2A, 2B, and 2C separately) and pooled population (2A and 2B combined).

2.1 Primary Endpoint(s)

2.1.1 Phase 1

Primary endpoints in Phase 1 are:

- Occurrence of DLTs.
- Frequency, duration, and severity of treatment-emergent adverse events (TEAEs), treatment-related TEAEs (TRAEs), and SAEs.
- Incidence and shifts of clinically significant clinical laboratory abnormalities.
- Change from baseline in other observations related to safety, including electrocardiograms, vital signs, ophthalmologic examination findings, and performance status.
- PK parameters (parent and relevant metabolites): maximum plasma concentration (C_{max}), time to maximum plasma concentration (T_{max}), area under the plasma concentration versus time curve from time 0 to t (AUC_{0-t}), area under the concentration versus time curve from time 0 to 24 hours (AUC_{0-24}), apparent oral clearance (CL/F), apparent volume of distribution (V_z/F) and half-life ($t_{1/2}$).

2.1.2 Phase 2

Primary endpoints in Phase 2 are:

- CR+CRh rate
- Frequency, duration, and severity of treatment-emergent adverse events (TEAEs), treatment-related TEAEs (TRAEs), and SAEs.
- Incidence and shifts of clinically significant laboratory abnormalities.
- Change from baseline in other observations related to safety, including electrocardiograms, vital signs, ophthalmologic examination findings, and performance status.

2.2 Secondary Endpoint(s)

2.2.1 Phase 1

Note that no secondary endpoints have been defined for Phase 1.

2.2.2 Phase 2

Secondary endpoints in Phase 2 are:

- Transfusion independence, defined as any transfusion-free period lasting for 56 consecutive days during which the patient is either on SNDX-5613 therapy or after cessation of SNDX-5613 therapy but prior to the start of new therapy.
- CRc rate (ie, CR+CRh+CRi+CRp).
- ORR (CRc+MLFS+PR).
- TTR.
- DOR
- EFS.
- OS.
- PK parameters: C_{max} , T_{max} , AUC_{0-t} , AUC_{0-24} , CL/F, V_z/F and $t_{1/2}$.

2.3 Exploratory Endpoint(s)

2.3.1 Phase 1

The exploratory endpoints in Phase 1 are:

- CR+CRh rate.
- Response relationship with correlative biomarkers in blood and bone marrow

2.3.2 Phase 2

The exploratory endpoint in Phase 2 is:

- Response relationship with correlative biomarkers in blood and bone marrow.

2.4 Safety Endpoint(s)

Please see Section 2.1 Primary Endpoints.

2.5 Food Effect

The exploratory food effect endpoints are:

- PK parameters for SNDX-5613 and SNDX-60165 (and any relevant metabolites): C_{max} , T_{max} , AUC_{0-t} , AUC_{0-24} , CL/F, V_z/F , $t_{1/2}$.

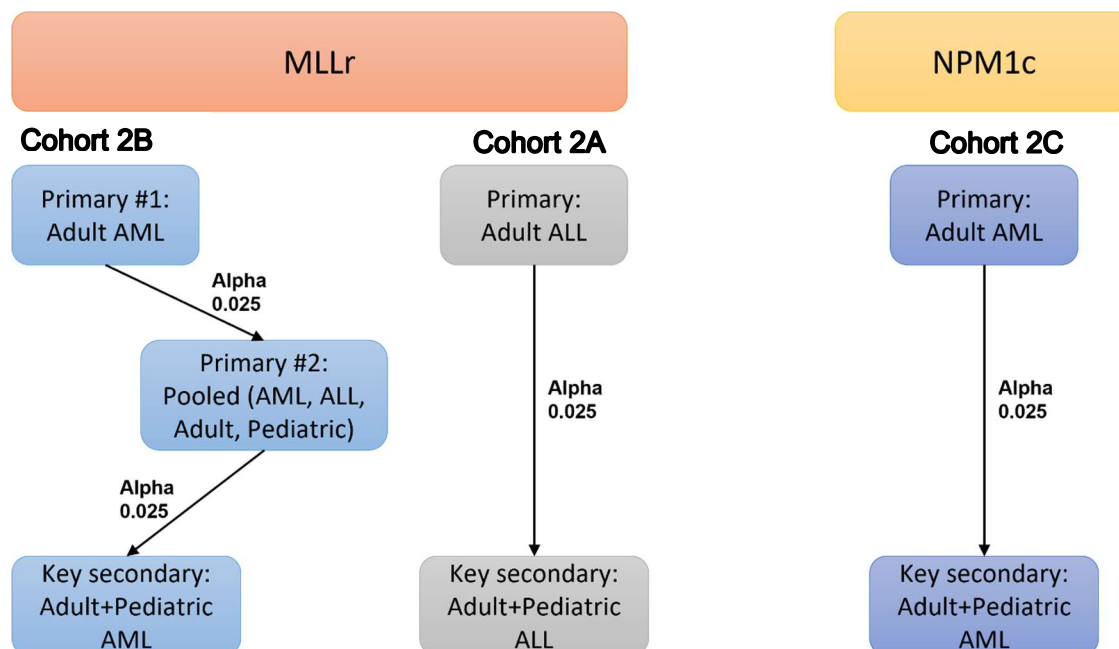
3 SAMPLE SIZE AND POWER

There is no formal hypothesis testing in Phase 1. However, a dose cohort in either arm may be expanded to 12 patients (Section 3.1.1 in protocol).

There is a primary hypothesis (Section 9.1.1 in protocol) and a key secondary hypothesis (Section 9.1.2 in protocol) in each of the cohorts in Phase 2 ([Figure 1](#)).

In addition, there will be a primary hypothesis (primary # 2) for a pooled KMT2Ar population based on the combined Cohort 2A and Cohort 2B data, which will be tested at the at the same time as the analysis timepoints for Cohort 2B ([Figure 1](#)). This pooled population will include efficacy evaluable adult and pediatric patients with MLLr AML or ALL/MPAL, and the test will use the same null hypothesis of 10% CR/CRh rate. If the test for the hypothesis in the adult AML efficacy evaluable population is significant, the hypothesis for the pooled population will be tested at the 1-sided significance level of 2.5%. With this hierarchical testing procedure, the overall Type I error rate for testing these two primary hypotheses (ie, adult AML and pooled) will be controlled under 0.025.

Figure 1. Hierarchical Testing Procedure for Primary and Key Secondary Hypotheses



Abbreviations: ALL = acute lymphoblastic leukemia; AML = acute myeloid leukemia; MLLr = mixed lineage leukemia 1 gene; NPM1c = nucleophosmin 1 mutation.

The actual number of patients to be enrolled in Phase 1 is dependent on the dose level(s) at which toxicities are seen and the number of dose levels investigated to identify the MTD and RP2D. It is anticipated that up to 114 patients will be enrolled in Phase 1, with 30 in Arm A, 24 in Arm B, 18 patients in Arm C, 12 patients in Arm D, 18 patients in Arm E, and 12 patients in Arm F. Backfill patients will also be enrolled into the study; the number of backfill patients will depend on the number of cohorts opened to backfill. No formal sample size calculation was performed for Phase 1 of the study.

In Phase 2 of the study, the sample size was driven by the primary analysis in the efficacy evaluable population (See Section 4.5) in the adult patients (primary #1). The number of adult efficacy evaluable patients in each cohort evaluated in each stage and the minimum number of responders needed to continue to the next stage are determined based on the minimax version of Simon's 2-stage design (Simon 1989), with 90% power and 1-sided significance level of 2.5%. Sixty-four (64) adult efficacy evaluable patients will be enrolled in each cohort. A true CR+CRh rate of 25% is hypothesized. A CR+CRh rate greater than 10% is considered the lower threshold for antileukemic activity. Based on the design elements specified above, up to 38 adult efficacy evaluable patients will be enrolled in each cohort during the first stage: If 5 or more adult evaluable patients achieve CR+CRh, 26 additional adult evaluable patients will be enrolled during second stage. Otherwise, enrollment will be terminated for that cohort. Upon completion of the second stage, if 12 or more patients out of 64 enrolled adult evaluable patients in a cohort achieve CR+CRh, the study will be considered positive. If the true CR+CRh rate is 10% or less for a leukemic subtype, the probability of terminating enrollment at the end of the first stage is 67%. Back-fill patients treated at RP2D and who were enrolled after 23 September 2021 will be included in the first stage of the Simon 2 stage design.

For Cohort 2B, a superiority/efficacy boundary will be added at the time of the interim analysis. Therefore, there will be both futility and efficacy analyses at the interim and a final analysis of efficacy for this cohort. The efficacy boundary was computed using package gsDesign version 3.1.1 in R version 4.0.2. At the interim analysis when 38 adult AML efficacy evaluable patients have been enrolled and sufficiently followed (Protocol Section 9.5), if 10 or more adult AML evaluable patients achieve CR/CRh, the primary endpoint for Cohort 2B will be met and the study will be considered positive. In this case, the interim analysis will be considered as the primary analysis time point for Cohort 2B and no more formal statistical testing will be conducted on the primary endpoint for Cohort 2B. If not, Cohort 2B will continue unless the futility boundary is met (ie, ≤ 4 adult AML evaluable patients achieve CR/CRh). The addition of this efficacy boundary spends a small amount of alpha (<0.004) and preserves the overall alpha under 0.025.

The hypothesis for the KMT2Ar pooled population (Cohort 2A + Cohort 2B) will also be tested at the time of the interim analysis and final analysis for Cohort 2B. At the time of the interim analysis for Cohort 2B, approximately 58 efficacy evaluable patients are expected to be available

from the pooled population. At the time of the final analysis for Cohort 2B, approximately 102 evaluable patients are expected to be available from the KMT2Ar pooled population. The efficacy boundaries for the interim and final analyses in the KMT2Ar pooled population are ≥ 12 and ≥ 18 CR/CRh responders, respectively. If any of the efficacy boundaries is crossed, the test for CR/CRh rate in the pooled population will be considered statistically significant and the corresponding analysis will be considered the primary analysis time point. These boundaries were computed using package gsDesign version 3.1.1 in R version 4.0.2. The alpha spent at the interim analysis is 0.0112 and the overall alpha is under 0.025. The boundaries will be re-calculated using the actual enrollment numbers at the time of the analyses, with the same amount of alpha spending (approximately 0.01) at the interim.

After enrollment of 64 evaluable adult patients in any cohort, if additional safety, efficacy, or PK data are needed to characterize any formulation, up to approximately 35 additional patients may be enrolled, with centrally confirmed mutation status.

Each cohort of the Phase 2 study, which may contain both adult and pediatric patients uses a single proportion binomial test with about 96.5% power and 1-sided significance level of 2.5%. A maximum of 84 evaluable patients (adult plus pediatric) will be enrolled in each cohort. A true CR+CRh rate of 25% in the overall population is hypothesized. The CR+CRh $> 10\%$ in overall population is considered the lower bound for anti-leukemic activity in patients with R/R acute leukemia. If 15 or more patients with CR+CRh are observed among the 84 evaluable patients in a cohort, then the lower limit of 95% confidence interval is greater than or equal to 10%. Thus, a total of 287 evaluable patients are planned to be enrolled in Phase 2.

Approximately 12 patients are anticipated to be sufficient to provide a preliminary estimate of the magnitude and variability of the food effect. Descriptive summaries will be provided.

Therefore, it is anticipated that up to 401 patients (not including back-fill patients in Phase 1) are to be enrolled and treated with SNDX-5613 in this study.

Table 1: Number of patients for the primary hypotheses

	Primary Hypothesis in each cohort	Primary Hypothesis in KMT2Ar Pooled population (estimated based on enrollment projection)
First stage (n)	38	~58
Second stage (n)	26	~44

4 ANALYSIS POPULATIONS

4.1 ITT Population

ITT Population will include all patients enrolled who provide informed consent. This will be the analysis set for the patient disposition.

4.2 DLT Population

The DLT Population will include all phase 1 patients with data used for implementing the dose-escalation schedule. These patients will have received 75% of all study drug doses in C1 (i.e., the DLT evaluation period) or stopped treatment because of DLTs in the DLT evaluation period.

4.3 KMT2Ar Population

The KMT2Ar Population will include all Phase 1 patients who received at least one dose of study drug and who have a KMT2A Gene Rearrangement at baseline based on local lab.

4.4 NPM1c Population

The NPM1c Population will include all Phase 1 patients who received at least one dose of study drug and who have a Nucleophosmin 1 (NPM1) mutation at baseline based on local lab.

4.5 Efficacy Evaluable Population

The Efficacy Evaluable Population consists of all Phase 2 patients who have received at least one dose of study drug, have been centrally confirmed for mutational status, and have at least 5% blasts in bone marrow at baseline. Efficacy analyses in Phase 2 will be performed using the Efficacy Evaluable Population.

Peripheral blood collected prior to start of study treatment may be used in cases where the bone marrow aspirate sample for central mutational confirmation is unavailable.

4.6 Safety Population

The Safety Population will include all enrolled patients who received at least one dose of study drug during the study. This will be the analysis set for all safety evaluations. This analysis set is also used for the efficacy analysis in Phase 1.

4.7 Pharmacokinetic Population

The Pharmacokinetic Population will include all patients who receive at least one dose of SNDX-5613 and have at least one valid plasma concentration of SNDX-5613 determined. This will be the analysis set for all PK analysis.

4.8 Formulation Effect Population

The Formulation Effect Population will include all patients who receive SNDX-5613 under tablet and capsule conditions and have sufficient PK data to calculate $C_{\max}$ or AUC_{0-t} and AUC_{0-12} following both conditions. This will be the analysis set for formulation effect analysis.

4.9 Food Effect Population

The Food Effect Population will include all patients who receive SNDX-5613 under fed and fasted conditions and have sufficient PK data to calculate $C_{\max}$ or AUC_{0-t} and AUC_{0-12} following both conditions. This will be the analysis set for food effect analysis.

4.10 Pharmacodynamic Population

The Pharmacodynamic Population will include patients who are exposed to SNDX-5613 and have sufficient samples collected to permit pharmacodynamic analyses.

4.11 Application of Analysis Population

Unless otherwise noted, the analysis populations that will be used for table summaries of each type is provided in the following table.

Table 2: Application of Analysis Populations

	ITT population	Safety population	Phase 2 Efficacy Evaluable population	Phase 1 KMT2Ar population	Phase 1 NPM1c population
Disposition	x	x	x	x	x
Baseline		x	x	x	x
Efficacy			x	x	x
Safety		x			

4.12 Protocol Deviations/Violations and Exclusions from Analysis Sets

A protocol deviation is defined as an instance of failure to follow, intentionally or unintentionally, the requirements of the protocol procedures or requirements. All violations and exclusions of patients from analysis sets if applicable will be identified at the Classification Meeting just prior

to the database lock and the final analysis, through clinical review input provided by Sponsor, using the Protocol Deviation Logs, provided by ICON Medical.

Further, deviations from protocol will be classified as major or minor.

Per Protocol (PP) analysis set will not be used, but major protocol deviations will be summarized in a summary table and all protocol deviations listed.

5 STATISTICAL CONSIDERATIONS AND ANALYSIS

5.1 Derived Variables

5.1.1 Derived General Variables

The below table (Table 3) provide the list of derived general variables for demographic and baseline characteristics, various duration derivations, baseline derivations applicable for this study.

Table 3: Derived General Variables

Variables	Formula
Demographic and Baseline characteristics	
Age at informed consent (in years)	Age = The number of years between Date of informed consent and Date of birth. If month is missing, use the 1 st of Jan in that year to impute the date of birth; If only the day of the month is missing, use the 1 st day of the month to impute the date of birth
Body mass index (BMI) (kg/m ²)	weight (kg)/[height (m)] ²
Derivation of Dosing	
Body surface area (BSA)	Patients weighing < 40kg will have their BSA calculated at the beginning of each cycle as $\sqrt{\frac{\text{Weight (kg)} * \text{Height (cm)}}{3600}}$ If there is a change of BSA $\geq 10\%$ from previous calculation, the planned dose should be recalculated based on the new BSA.
Study Day	
Study day at any visit	Date of interest – date of first dose of study drug. One day is added if this difference is ≥ 0
Baseline Derivations	
Baseline	Latest non-missing measurement prior to first dose of study medication
ECG result	All ECG recordings are taken in triplicate. The mean of the 3 values are used in the summaries. Baseline is defined as the mean of the predose triplicate measurements obtained on Cycle 1 Day 1, or, if missing,

	the mean of the triplicate measurements obtained at the screening visit. If less than three measurements are available, then the mean of the available measurements are used.
Change from baseline	Test value at visit X – Baseline value
Percentage change from baseline	$(\text{Test value at visit X} - \text{Baseline value}) / \text{Baseline value} \times 100\%$
Adverse event duration (in days)	Event end date - Event onset date + 1.

5.1.2 Derived Efficacy Endpoints

5.1.2.1 CR + CRh rate

CR + CRh rate is defined as the percentage of patients with at least one visit response of CR or CRh defined as in Section 8.1.2 of the protocol which is consistent with the definition in FDA AML guidance. Investigator reported visit response (including response assessment and date of response) will be used for the primary analysis of CR+CRh rate. Only data up to the start of new anti-leukemic therapy will be included in the assessment of CR +CRh rate. Patients who attain a best response of CRp or CRi prior to transplant and then proceed to transplant without receiving additional anti-leukemic therapy and have hematologic improvement post-transplant to meet criteria for response of CR or CRh post-transplant will be counted as responders for CR+CRh. Patients who do not have any baseline or post-baseline disease assessment will be counted as non-responders.

5.1.2.2 CRc rate (CR+CRh+CRi+CRp).

CRc rate is defined as the percentage of patients with at least one visit response of CR, CRh CRi or CRp defined as in Section 8.1.2 of the protocol which is consistent with the definition in FDA AML guidance. Investigator reported visit response (including response assessment and date of response) will be used for the primary analysis of CRc rate. Only data up to start of new anti-leukemic therapy will be included in the assessment of CRc rate. Patients who do not have any baseline or post-baseline disease assessment will be counted as non-responders.

5.1.2.3 ORR

ORR is defined as the percentage of patients with at least one visit response of CR, CRh CRi, CRp, MLFS, or PR defined as in Section 8.1.2 of the protocol which is consistent with the definition in FDA AML guidance. Investigator reported visit response (including response assessment and date of response) will be used for the primary analysis of ORR. Only data up to

the start of new anti-leukemic therapy will be included in the assessment of ORR. Patients who do not have any baseline or post-baseline disease assessment will be counted as non-responders.

5.1.2.4 Minimal residual disease (MRD) negative status

If there is a MRD negative assessment at any visit on or after the first CRc response and up to the start of new anti-leukemic therapy, the patient is considered as MRD negative.

5.1.2.5 Transfusion independence

Transfusion independence, defined as any transfusion-free period lasting at least 56 consecutive days, during which the patient is either on SNDX-5613 therapy or after cessation of SNDX-5613 therapy but prior to the start of new therapy. The start of the transfusion free period can be on or after the first dose of SNDX-5613 treatment, and up to one day after the last dose of treatment or the start of HSCT and conditioning regimen or new anti-leukemic therapy, whichever is earlier. The end of the transfusion free period has to be before the start of new therapy (excluding HSCT and conditioning regimen and treatment post-HSCT with study drug or other maintenance treatment). Patients who died or started new anti-leukemic therapy (including HSCT and conditioning regimen) before having a transfusion-free period lasting for at least 56 consecutive days will not be considered as transfusion independent.

5.1.2.6 Event-Free Survival (EFS)

EFS is defined as the number of months from the date of first dose to the date of first documented relapse/progression or death, whichever occurs first. The censoring rules are summarized in Table 4.

Table 4. Censoring Rules for EFS

Scenario	Censoring Date
No post-baseline assessment, no death	Date of first dose
No relapse/PD or death	Date of the last non-missing/non-unknown response assessment
Two or more consecutive missing scheduled response assessments from last response assessment to documented relapse/PD or death	Date of last non-missing/unknown response assessment prior to missed scheduled assessments (this rule does not apply to post-transplant response assessments; all response assessments post-transplant will be included in the calculation)

Given the scheduled visit assessment schedule (i.e. every 4 weeks (± 7 days)), for on treatment

visits, two missing visits will equal to 10 weeks since previous response assessment, allowing for early and late visits.

5.1.2.7 Duration of Response (DOR)

DOR will be calculated for patients who achieve a CR/CRh or CRc. DOR is defined as the date of first documented response to the first documented relapse or death. For CRc, DOR is defined as the date of first documented CR, CRh, CRi or CRp to the first documented relapse or death. Investigator reported date of response will be used in the primary analysis of DOR. DOR will be calculated for patients who achieve a CR, CRh, CRi or CRp. DOR will follow the censoring rules in Table 5.

Table 5. Censoring Rules for Duration of Response

Scenario	Censoring Date
No relapse/PD or death	Date of the last non-missing/non-unknown response assessment
New anti-leukemic therapy excluding transplant and associated preparative and conditioning regimen and post-transplant maintenance including SNDX-5613 after HSCT*, and excluding hydroxyurea/steroid use per protocol, prior to documented relapse/PD or death	Date of the last non-missing/non-unknown response assessment prior to anti-leukemic therapy
Two or more consecutive missing scheduled response assessments from last response assessment to documented relapse/PD or death	Date of last non-missing/non-unknown response assessment prior to missed scheduled assessments (this rule does not apply to post-transplant response assessments; all response assessments post-transplant will be included in the calculation)

* Includes resumption of SNDX-5613 after HSCT per protocol Section 6.7 and in other scenarios. Given the scheduled visit assessment schedule (i.e. every 4 weeks (± 7 days)), for on treatment visits, two missing visits will equal to 10 weeks since previous response assessment, allowing for early and late visits.

5.1.2.8 Time to Response (TTR)

TTR is defined as the number of months from the date of first dose to the date of initial response (CR/CRh or CRc).

TTR (months) will be calculated as $(\text{First initial response date} - \text{first dose date} + 1) / 30.4375$.

5.1.2.9 Overall Survival (OS)

OS is defined as the time from the date of first dose to the death due to any cause.

OS (months) will be calculated as $(\text{Death date} - \text{first dose date} + 1) / 30.4375$ if a patient died by cut-off date; or $(\text{date of last known survival follow-up} - \text{first dose date} + 1) / 30.4375$ when a patient is deemed to be censored. Patients known to be alive or lost to follow-up before data cut-off date will be censored at the last known survival follow-up date.

5.1.2.10 Duration of follow up

Duration of follow up is the period from the first dose date to the date of data cutoff date or study completion/discontinuation date, whichever occurs earlier.

For patients who had end of study date, duration of follow up (months) = $(\text{End of study date} - \text{first dose date} + 1) / 30.4375$; if no end of study date before data cutoff date, duration of follow up (months) = $(\text{data cutoff date} - \text{first dose date} + 1) / 30.4375$

5.2 Handling of Missing Data and Outliers

5.2.1 Missing Data Analysis Methods

Not applicable

5.2.2 Handling of Missing or Incomplete Dates

Initial disease diagnosis date

Partial date of initial disease diagnosis will be imputed as below:

- If year is missing (or completely missing), do not impute;
- 15th of the month will be used to impute the initial diagnosis date if only the day is missing;
- June 15st of the year will be used to impute as the initial diagnosis date if both the day and the month are missing.

The imputed date will be compared to reference start date (i.e. enrolled date). If it is later than the reference start date, the reference start date will be used to impute the incomplete date.

Imputation rules for missing or partial AE date are defined below:

If only Day of AE start date is missing:

If the AE start year and month are the same as that for the first dose date, then:

- If the full (or partial) AE end date is NOT before the first dose date or AE end date is missing, then impute the AE start day as the day of first dose date.
- Otherwise, impute the AE start day as 1.

Compare the imputed AE start date with Treatment Emergent (TE) period to determine whether the AE is pre-treatment AE or posttreatment

If Day and Month of AE start date are missing:

If AE start year = first dose year, then:

- If the full (or partial) AE end date is NOT before the first dose date or AE end date is missing, then impute the AE start Month and Day as the Month and Day of first dose date.
- Otherwise, impute the AE start MONTH as January and the DAY as 1.

Compare the imputed AE start date with Treatment Emergent (TE) period to determine whether the AE is pre-treatment AE, TEAE or post-treatment AE.

If Year of AE start date is missing:

If the year of start date is missing or AE start date is completely missing, then query site with no imputation. Also compare the partial AE end date to the first dose date. If the AE end date is before the first dose date, then the AE should be considered as a pre-treatment AE. Otherwise, the AE will be considered as TEAE.

Impute Missing AE Stop Dates

If the imputed stop date is before the AE start date then the imputed stop date will be equal to the start date.

Missing day and month

- If the year of the partial stop date is the **same** as the year of the last dose date of study medication, then the day and month of the last dose date will be assigned to the missing fields.
- If the year of the partial stop date is **prior to** the year of the last dose date of study medication, then December 31 will be assigned to the missing fields.
- If the year of the partial stop date is **after** the year of the last dose date of study medication, then January 1 will be assigned to the missing fields.

Missing day only

- If the month and year of the partial stop date are the **same** as the month and year of the last dose date of study medication, then the day of the last dose date will be assigned to the missing day.
- If the month and year of the partial stop date are **before** the month and year of the last dose date of the study medication, then the last day of the month will be assigned to the missing day.
- If the month and year of the partial stop date are **after** the month and year of the last dose date of study medication, then the first day of the month will be assigned to the missing day.

Imputation rules for missing or partial concomitant medication start/stop dates are defined below:

Missing or partial medication start date:

- If only DAY is missing, use the first day of the month.
- If DAY and Month are both missing, use a date/month before the first dose date/month or medication stop date/month whichever is earlier, or use June 15th of the year if both the reference date/month are missing
- If DAY, Month and Year are all missing, use a date/month/year before the first dose date/month/year or medication stop date/month/year whichever is earlier.

Missing or partial medication stop date:

- If only DAY is missing, use the last day of the month.
- If DAY and Month are both missing, use the last day of the year.
- If DAY, Month and year are all missing, assign 'continuing' status to stop date

Impute Missing Dates in Prior Anti-leukemic Regimen

For each prior anti-leukemic regimen, the regimen start/stop date, disease progression date, best response date, and date of first response for PR or better, will be collected. If the day of the start date is missing, then the first day of the non-missing month will be assigned to the missing day; if month or year of the date is missing, the date will not be imputed and treated as missing.

Impute Missing Dates in Subsequent Anti-leukemic Therapy

Patients will be allowed to take other anti-leukemic therapy after being discontinued from the study treatment. The anti-leukemic therapy start date will be collected. If the day of any date is missing, then the first day of the non-missing month but not earlier than the last dose of study drug will be assigned to the missing day.

6 STATISTICAL METHODS

6.1 General Statistical Conventions

All statistical procedures will be completed using SAS version 9.4 or higher. Descriptive statistics will be presented for all the endpoints by cohort unless specified otherwise.

The safety and efficacy analyses will be presented by study phase. For Phase 1, tabulations will be provided by dose level and overall. For Phase 2, tabulations will be provided by leukemia cohort and overall if applicable.

Confidence intervals (CIs) will be presented as 2-sided 95% CIs unless specified differently in specific analysis.

Categorical variables will be summarized using number (n) and percentage of in each category including missing category if any. All percentages will be rounded to two digits after decimal point. The number and percentage will be presented in the form XX (XX.XX %), where the percentage is in the parentheses. If the percentage of a category is “100 %”, then display it as “100 %” (do not add decimal after 100).

Continuous variables will be summarized using number of patients (n), mean, median, standard deviation (SD), minimum, and maximum for continuous variables. All mean and median values will be formatted to one more decimal place than the measured value. Standard deviation values will be formatted to two more decimal places than the measured value. The minimum and maximum will be displayed to the same number of decimal places as the original data.

Time-to-event data will be analyzed using the Kaplan- Meier method and results will be summarized by 25th, 50th (median) and 75th percentiles with associated 2-sided 95% CIs, as well as the percentage of censored observations.

Unless otherwise noted, baseline is defined as the last non-missing value recorded prior to the first dose of trial drug.

All patient data will be presented in individual patient data listings. In general, no imputed dates or values will be presented in listings unless specified.

6.2 Patient Disposition

The number of patients included in each analysis set will be summarized by arm/dose in Phase 1 and by cohort in Phase 2, along with the reason for any exclusions. Patients discontinuing from study treatment and/or withdrawing from study participation the primary reason for discontinuation will be summarized by arm/dose in Phase 1 and by cohort in Phase 2. The number of patients who resumed study treatment post-HSCT may be summarized along with the reason for discontinuing from post-HSCT study treatment.

A listing of patient disposition will be provided for all ITT Population, with the extent of their participation in the study and the reason for discontinuation.

6.3 Protocol Deviations

A summary of major protocol deviations occurred will be tabulated by arm/dose in Phase 1 and by cohort in Phase 2 for Safety Population.

All protocol deviations will be listed.

6.4 Demographics and Baseline Characteristics

6.4.1 Demographics

Background and demographic characteristics of the Safety Population and Efficacy Evaluable Population including age, BSA, sex, ethnicity, race, height, weight, and BMI will be summarized by arm/dose in Phase 1 and by cohort in Phase 2 using descriptive statistics or frequency tables. Categories will be presented for different age groups, BMI groups (Severely underweight, Underweight, Normal weight, Overweight and Obesity).

Listings of demographics will be produced for Safety Population.

6.4.2 Baseline Characteristics

Baseline characteristics of the Safety Population and Efficacy Evaluable Population including:

- Disease Diagnosis: including leukemia type, time from date initial diagnosis to date of enrolled, WHO leukemia type, extramedullary disease, site(s) of extramedullary disease, MLL rearrangement, NPM1 Mutation, IDH1 Status, IDH2 Status, DNMT3A, FLT
- Prior Leukemia Therapy: total number, categories of 1, 2, >= 3 lines and summary statistics of prior anti-leukemia regimens prior HSCT transplant, prior venetoclax, best response on most recent leukemia therapy, outcome.
- Bone marrow status
- Previous treatments: including radiation therapy, and response to such treatment
- ECOG, Karnofsky or Lansky performance status
- ECHO/MUGA
- Transfusion dependence status
- Disease status at baseline

Time from date of primary diagnosis to date of enrolled = (Date of enrolled - date of the first disease diagnosis + 1) / 30.4375

Baseline characteristics will be summarized by arm/dose in Phase 1 and by cohort in Phase 2 using descriptive statistics or frequency tables.

Listings of baseline characteristics will be produced for Safety Population.

6.4.3 Medical History

Medical history from the Medical History eCRF page will be coded with medical dictionary for regulatory activities (MedDRA) version 19.0 or above. All medical histories will be listed and the number and percentage of patients with any medical history will be summarized for the Safety Population by primary system organ class (SOC) and preferred term (PT).

6.4.4 Prior and Concomitant Medications

Incomplete medication start and stop dates will be imputed as detailed in [Section 7.2.2](#). Medications used in this study will be coded by using the latest available version of the World Health Organization Drug Dictionary Enhanced (Enhanced 2019 March version, or higher).

Prior medications: A medication that starts prior to the date and time of first dose of study medication

Concomitant medications: A medication that is ongoing or starts on or after the time of the first study drug dose through 30 days after the last dose of the study drug.

If a medication starts before the date of first dose of study medication and is ongoing (i.e., does not stop before) the date of first dose of study medication, then the medication will be counted as both a prior and a concomitant medication.

If medication start and/or stop dates and times are missing or partial, the dates will be compared as far as possible with the date and time of first dose of study medication. A medication will be assumed to be prior unless there is clear evidence (through comparison of partial dates and times) to suggest that the medication stopped before or started after the first dose of study medication. A medication will be assumed to be concomitant unless there is clear evidence (through comparison of partial dates and times) to suggest that the medication stopped prior to the first dose of study medication.

Prior and Concomitant medications will be summarized for the safety analysis set by anatomical therapeutic chemical (ATC) level 2 and preferred term.

Listing for medications data will be produced for Safety Population. A listing for the concomitant medications known to cause prolonged QTc will be produced for Safety Population.

6.5 Extent of Exposure

Summary of exposure data will be presented by arm/dose in Phase 1 and by cohort in Phase 2 for safety analysis set.

Listings of all doses of the study drug along with dose change reasons will be produced.

6.5.1 Exposure

Overall duration of treatment (days) = (Date of last administration of study drug – Date of first administration of study drug) +1.

The overall duration of study treatment administration and number of cycles initiated will be summarized for the safety analysis set. The total number of cycles started and the total number of cycles started by category (<1 , ≥ 1 , ≥ 2 , ≥ 3 , so on) will be tabulated for each patient and summarized for dose cohort/tumor type. The relative dose intensity of SNDX-5613-0700 administered will be calculated. These data will be further summarized by calculating the mean, standard deviation, median, and range of these values. The number and proportion of patients with 1 or more dose modifications (i.e., dose reduction, re-escalation and missed dose) will also be tabulated and listed along with the reason for modification.

For patients who resume treatment after HSCT, the summary will include both the period before HSCT and the period after HSCT. Additional summaries for the period after HSCT may be performed.

6.5.2 Treatment Compliance

- Overall actual cumulative dose in the whole treatment = Sum of (actual dose taken (in mg) for each dosing day during the whole treatment period).
- Overall assigned cumulative dose in the whole treatment = Sum of (assigned dose (in mg) for each dosing day during the whole treatment period).
- Relative dose intensity = (overall actual cumulative dose / overall assigned cumulative dose) x 100%.

The relative dose intensity will be summarized by using standard summary statistics for the safety analysis set.

6.6 Efficacy Analyses

For Phase 1, statistical analyses will be descriptive and graphical in nature. No formal statistical hypothesis testing will be performed. Efficacy analysis will be summarized by arm/dose. In addition, RP2D dose levels may be combined for some summary tabulation.

For each cohort in Phase 2, the adult efficacy evaluable population will be the primary analysis set for the efficacy analyses. The analysis of these endpoints will be repeated using overall (adult plus pediatric) efficacy evaluable population for each cohort. The efficacy evaluable patients in the pooled population will be the primary analysis set for the efficacy analysis in the pooled population (adult and pediatric patients in AML or ALL/MPAL). Analyses of the pediatric population only will be performed with descriptive statistics. Analyses of ALL/MPAL patients from the pooled population will be performed with descriptive statistics. Additional patients that may be enrolled after enrollment of 64 evaluable adult patients will not be included in the primary analysis set for the efficacy analyses.

The SAS code is presented in [Appendix 1](#).

Listing for efficacy data including response assessments post-transplant will be produced for Safety Population, the listing will be provided for Phase 1 and Phase 2 separately.

6.6.1 Analysis Methods

6.6.1.1 Analyses of Time-to-Event Endpoints

Time-to-event endpoints will be summarized using the Kaplan-Meier method and displayed graphically when appropriate. Estimated event time at 3, 6, 12 and 24 months and median event times with 2-sided 95% confidence intervals will also be provided when appropriate.

6.6.1.2 Analyses of Binary Endpoint

The rates of binary endpoints will be provided along with the corresponding 2-sided confidence intervals using an exact method.

6.6.2 Analysis of Primary Efficacy Endpoint(s)

6.6.2.1 CR+CRh rate

- For Phase 1, the analysis for CR rate (CR+CRh) will be performed within KMT2Ar population and NPM1c population.
- For Phase 2, the analysis for CR+CRh rate will be performed within the Efficacy Evaluable population. The proportion along with 95% exact binomial confidence interval will be presented.
- Sensitivity analysis will be performed using a defined window applied for the CBC that is used in conjunction with the bone marrow to determine response assessment at each visit. All CBC testing within -7 to + 15 days from the date of the bone marrow will be used to derive the response assessment at each visit, with 15 days defined as up to Cycle 2 Day 15 visit, or up to 15 days from the Day 1 bone marrow exam in Cycle 3 and beyond. Date of bone marrow used to establish response will be considered as the date of response.
- Sensitivity analysis will be performed to evaluate the impact of hematopoietic growth factors and platelet transfusions. CBC with platelet transfusion or hematopoietic growth factor within 7 days prior won't be used to support the response assessment of CR/CRh in this sensitivity analysis.
- Sensitivity analysis will be performed to provide an unconditional uniformly minimum variance unbiased estimator for CR rate ([Jung 2004](#)).

- Best overall response and MRD negativity rate will be summarized as supportive analysis.
- Listings will be provided for patients who received hematopoietic growth factors and platelet transfusions within 7 days prior to the response assessment.

6.6.3 Analysis of Secondary Efficacy Endpoint(s)

6.6.3.1 CRc (CR+CRh+CRi+CRp) rate and ORR

The CRc (CR+CRh+CRi+CRp) rate and ORR analysis will be performed within KMT2Ar population, and NPM1c population for Phase 1 and Efficacy Evaluable Populations in phase 2. The proportion along with 95% exact binomial confidence interval will be presented by arm/dose in Phase 1 and by cohort in Phase 2.

6.6.3.2 Event-Free Survival (EFS)

The definition of EFS can be found in section [5.1.2.1](#). The analysis of EFS will be summarized by arm/dose in Phase 1 and by cohort in Phase 2.

The number of patients with an event (disease progression/relapse), number of patients censored, and reason for censoring will be summarized.

Summary statistics based on Kaplan-Meier method will be provided as below:

- Quartiles (25%, 50% and 75%) and 95% CI
- Event-free survival rate at time points 3, 6, 12 and 24 months.

The Kaplan-Meier curves may be displayed by arm/dose in Phase 1 if appropriate, and by cohort in Phase 2. Number of patients at risk at given points will be also included into the plot.

6.6.3.3 Duration of Response (DOR)

The definition of DOR can be found in section [5.1.2.2](#). DOR will be calculated for patients who achieve a CR/CRh or CRc. DOR will be summarized by arm/dose in Phase 1 and by cohort in Phase 2.

Sensitivity analysis will be performed using response assessment in the sensitivity analysis for CR+CRh rate following the same censoring rule as in the primary analysis for DOR.

Summary statistics based on Kaplan-Meier method will be provided as below:

- Quartiles (25%, 50% and 75%) and 95% CI

6.6.3.4 Time to Response (TTR)

The definition of TTR can be found in section [5.1.2.3](#). The analysis of TTR will be summarized by arm/dose in Phase 1 and by cohort in Phase 2.

6.6.3.5 Overall Survival (OS)

The definition of OS can be found in section [5.1.2.4](#). The analysis of OS will be summarized by arm/dose in Phase 1 and by cohort in Phase 2.

The number of deaths and the reasons for OS censoring (alive, lost-to-follow-up or withdrew consent) will be summarized.

Summary statistics based on Kaplan-Meier method will be provided as below:

- Quartiles (25%, 50% and 75%) and 95% CI
- Survival rate at time points 3, 6, 12, and 24 months.

The Kaplan-Meier curves may be displayed by arm/dose in Phase 1 if appropriate, and by cohort in Phase 2. Number of patients at risk at given points will be also included into the plot.

6.6.3.6 Transfusion Independence

The proportion of patients that are transfusion independent will be calculated separately for those who are transfusion dependent at baseline and for those who are not transfusion dependent at baseline along with exact 95% binomial CI. Transfusion dependence at baseline is defined as transfusions of red blood cells (RBCs) or platelets within 56 days prior to the first dose of treatment. The analysis will be performed by best overall response categories. Similar analysis will be repeated for RBCs and platelets separately. A figure of hemoglobin and platelet counts at each scheduled visit will be performed as a supportive analysis. A listing of growth factor usage will be provided and summary may be provided if appropriate.

6.6.3.7 Subgroup Analyses

Subgroup analyses will be performed for CR/CRh rate and DOR if appropriate.

These subgroup analyses will be carried out using the subjects from the Efficacy Evaluable Population.

The list of potential subgroups (with applicable definitions in parentheses) includes, but is not necessarily limited to, the following:

Age

- ≥ 18 vs. <18
- <18 , $18-<65$, ≥ 65

Sex

Race (White vs non-White)

Baseline ECOG (0 vs 1 vs 2)

Secondary AML (Yes vs No)

Prior HSCT (Yes vs No)

Primary Refractory (Yes vs No)

Number of prior lines of therapy (1 vs 2 vs ≥ 3)

Prior Venetoclax (Yes vs No)

CNS status

The following subgroups only apply to the pooled population:

Histology (AML vs ALL vs other)

6.7 Safety Analyses

Safety analyses set will be performed using safety population, unless otherwise specified. No formal statistical hypothesis testing will be conducted. DLT will be analyzed for Phase 1 only, and other safety endpoints will be analyzed for Phase 1 and Phase 2 separately.

The safety assessment will be based on AEs, laboratory values, vital signs, physical examination, ECGs, ophthalmologic examinations and ECOG, Karnofsky or Lansky performance status.

The safety analyses involving changes from baseline to a specific time point in safety variables (e.g., laboratory parameters, vital signs, and ECG) will only include patients from the safety population who have data available for both the baseline and the time point under consideration unless otherwise specified.

6.7.1 Dose Limiting Toxicity

The dose limiting toxicities will be summarized for dose escalation phase (Phase 1) by arm/dose using the safety population. The table will include number and percentage of patients with at least one DLT.

The listing will be listed by each arm and patient for the Safety population.

6.7.2 Adverse Events

AEs will be coded using the standard Medical Dictionary for Regulatory Activities (MedDRA), version 19.0 or higher.

All AEs will be graded according to the CTCAE version 5.0. The CTCAE displays Grade 1 through 5 with unique clinical descriptions of severity for each AE based on these general guidelines:

Grade 1 = Mild	Asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated.
Grade 2 = Moderate	Minimal, local or noninvasive intervention indicated; limiting age-appropriate instrumental activities of daily living.
Grade 3 = Severe or medically significant but not immediately life-threatening	Hospitalization or prolongation of hospitalization indicated; disabling; limiting self-care activities of daily living
Grade 4 = Life-threatening consequences	Urgent intervention indicated
Grade 5 = Death	Death related to AE

If a patient experiences the same AE more than once with different toxicity grades, then the event with the highest severity grade and strongest causal relationship to study drug will be tabulated in “by grade” tables. If a patient experiences multiple AEs under the same preferred term (system organ class), then the patient will be counted only once for that preferred term (system organ class). In addition, AEs with a missing intensity will be presented in the summary table as an intensity category of “Missing.”

Treatment-emergent AEs are defined as having onset on or after first study drug dosing date or a sign, symptom, or diagnosis that worsens on or after first study drug dosing date and no later than 30 days after the date of last study drug dosing.

For patients that undergo HSCT without leaving the study and plan to resume SNDX-5613 treatment after HSCT, TEAE is defined as AEs observed after starting administration of the study treatment until the last dose before HSCT plus 30 days, and AEs that begin after resumption of SNDX-5613 post HSCT and within 30 days after the last dose of SNDX-5613 will also be counted as TEAE.

For patients that undergo HSCT without leaving the study and plan to resume SNDX-5613 treatment after HSCT, the primary analyses for AE will be based on the period prior to HSCT, i.e. AEs observed after starting administration of the study treatment until the last dose plus 30 days or before the first day of HSCT and conditioning regimen, whichever is earlier. AEs reported during resumption of SNDX-5613 post HSCT may be summarized separately.

AEs and SAEs reported during HSCT while off SNDX-5613 as well as before resumption of SNDX-5613 will be listed.

The causal relationship between the occurrence of an AE and study drug is judged by the Investigator on the basis of his or her clinical judgment. The AE is reported by investigators “Related”, “Possibly Related”, “Probably Related”, “Unlikely Related” and “Not Related”. Relationship would be summarized as “Related” and “Not Related”. Adverse events with investigator indicated drug-event relationship of “Unlikely Related” and “Not Related” are classified as “Not Related” with the study medication, while adverse events with drug-event relationship documented as “Related”, “Possibly Related” and “Probably Related” are considered as “Related” with the study medication. TEAEs with a missing relationship to study medication will also be considered as “Related” with the study medication. If a patient reports the same AE more than once within that SOC/PT, the AE with the strongest relationship to study medication will be used in the corresponding relationship summaries.

TEAEs leading to Death are those events with “Outcome” response recorded as “Fatal” on the Adverse Events page of the eCRF.

Serious adverse events are those events with a response of “Yes” for the item “Was AE Serious” on the AE form.

Actions taken to study drug are dose discontinuation, dose reduced and dose interruption from eCRF page of Adverse Events.

An overview table (including only treatment-emergent AEs) will be also presented with the number (and percentage) of patients with:

- At least one TEAE
- At least one treatment-related TEAEs (TRAEs)
- CTCAE grade 3 or higher TEAEs
- CTCAE Grade 3 or higher treatment-related AEs
- At least one SAE
- Treatment related SAEs
- TRAEs leading to dose modifications (i.e. dose reduced or dose interrupted)
- TRAEs leading to dose reduction
- TRAEs leading to dose interruption
- TEAEs leading to study drug discontinuation
- TRAEs leading to study drug discontinuation
- TEAEs leading to death
- TRAEs leading to death.
- AEs of special interest (AESIs)

The number of events will be also included in this overview table.

The incidence rate for all TEAEs will also be summarized as below:

- TEAEs by SOC, PT.
- TEAEs by PT
- CTC grade 3 or higher TEAEs by SOC, PT
- CTC grade 3 or higher TEAEs by PT and CTCAE grade
- Treatment-related TEAEs by SOC, PT
- Treatment-related TEAEs by PT
- CTC grade 3 or higher treatment-related TEAEs by SOC, PT
- CTC grade 3 or higher treatment-related TEAEs by PT and CTCAE grade
- Serious TEAEs by SOC, PT
- Serious TEAEs by PT
- Treatment Related Serious TEAEs by SOC, PT
- Treatment Related Serious TEAEs by PT
- TEAEs leading to drug interruption by PT
- TEAEs leading to drug reduction by PT
- TEAEs leading to study drug discontinuation by PT
- Treatment-related TEAEs leading to drug reduction by PT
- TEAEs leading to Deaths, by PT
- TRAEs leading to Deaths, by PT

Listings will be provided as below:

- AEs
- AEs leading to Deaths
- Serious AEs
- AEs leading to dose interruption, reduction and discontinuation
- Death

6.7.2.1 TEAEs of Special Interest

TEAEs of special interest (AESI) as defined in the protocol are as following:

- Differentiation Syndrome
- Prolonged QTc
- Peripheral Neuropathy

Treatment emergent AESI and Serious AESI will be summarized by AESI category preferred term and CTCAE grading. Time to onset of prolonged QTc will be summarized. Listings of AESI will be provided.

Details on additional analysis for differentiation syndrome will be provided in a separate document.

6.7.3 Clinical Laboratory Analysis

All laboratory data will be summarized in International System of Units (SI). Quantitative laboratory measurements reported as “< X”, i.e. below the lower limit of quantification (BLQ), or “> X”, i.e. above the upper limit of quantification (ULQ), will be converted to X for the purpose of quantitative summaries, but will be presented as recorded, i.e. as “< X” or “> X” in the listings. The summaries will include all laboratory assessments collected no later than 30 days after the last drug/exposure date.

Descriptive statistics of absolute value and change from baseline (mean, SD, median, min, and max) for continuous assessments will be provided for minimum and maximum post-baseline assessments.

All laboratory assessments will be listed with values flagged to show corresponding CTC grades if applicable and abnormality assessed by investigator [normal/abnormal, not clinically significant (NCS)/abnormal, clinically significant (CS)].

6.7.3.1 CTCAE Grading for Laboratory Data

Laboratory data as below will be classified into CTC grades according to the NCI Common Terminology Criteria for Adverse Events (CTCAE) v5.0.

- Hematology: WBC count with differential, platelet count, hemoglobin, hematocrit
- Biochemistry: ALT, AST, albumin, alkaline phosphatase, total and direct bilirubin, blood urea nitrogen, calcium, creatinine, sodium, potassium, chloride, bicarbonate, glucose, lactate dehydrogenase (LDH), phosphorus, total protein, uric acid, magnesium (baseline only, unless clinically indicated)

A severity grade of 0 will be assigned when the value is within normal limits.

Some modifications to the grading system will be applied:

- Grade 5 refers to fatal outcomes, which cannot be determined solely by lab values, therefore will not appear in the grading system.
- A further category denoted Grade 0 would include all other laboratory values except

missing values.

- Missing results shall be graded as missing.

The following summaries will be produced for the laboratory data (by laboratory parameter):

- Shift tables using CTC grades to compare baseline to the worst post-baseline value will be produced for hematology and biochemistry laboratory parameters with CTC grades.

For some specific parameters with CTCAE grading in both high and low direction (e.g., calcium, glucose, magnesium, potassium, sodium), CTCAE in high and low directions will be presented separately, i.e. hyper for higher values of concern and hypo for lower values of concern.

6.7.3.2 Laboratory Reference Ranges High/Low Criteria and Investigator's Assessment

Quantitative laboratory measurements will be compared with the relevant laboratory reference ranges in SI units and categorized as:

- Low: Below the lower limit of the laboratory reference range.
- Normal: Within the laboratory reference range (upper and lower limit included).
- High: Above the upper limit of the laboratory reference range.

In addition to the high and low quantitative laboratory assignments (as identified by means of the laboratory reference ranges), quantitative safety (and other) laboratory assessments will also be evaluated as normal/abnormal, not clinically significant (NCS)/abnormal, clinically significant (CS) by investigator.

The following summary table will be produced for lab tests that are not gradable by CTCAE (by laboratory parameter):

- Shift table from baseline to worst post-baseline value according to abnormality assessed by investigators (normal, CS and NCS). Values from post-baseline unscheduled visit will be considered when derive worst post-baseline value.

Summary of liver laboratory parameters related to potential Hy's law pattern will also be provided.

6.7.4 Vital Signs

Vital sign assessments are performed in order to characterize basic body function. The parameters collected are the following: systolic and diastolic blood pressure (mmHg), pulse (beats/min), and respiratory rate (breaths/min), as well as oxygen saturation and temperature (°C).

Descriptive statistics of actual value and change from baseline will be presented each scheduled visit for each parameter. Maximum and minimum post-baseline value and change from baseline will be also presented. Values from post-baseline unscheduled visits will be considered when deriving maximum and minimum post-baseline value.

Vital sign assessments will be presented for each patient in a data listing.

6.7.5 Physical Examinations

General physical examination assessments will be listed for each patient.

6.7.6 Electrocardiograms

The following ECG parameters will be reported for this study:

- RR Interval (msec)
- PR Interval (msec)
- QRS Interval (msec)
- QT Interval (msec)
- QTcF Interval (msec) = $QT/(RR^{0.33})$, where RR = 60 / heart rate (bpm)
- Heart Rate (beats/min)
- Overall assessment of ECG:
 - Normal
 - Abnormal , NCS
 - Abnormal , CS

The following summaries will be provided for ECG data:

- Descriptive statistics of actual value and change from baseline in ECGs will be presented by each scheduled visit.
- Descriptive statistics of maximum QTc and maximum changes from baseline will be presented. Values from post-baseline unscheduled visit will be considered when derive maximum QTc from baseline.
- The number and percentage of patients in each QTcF category (<450 msec, 450–480 msec, 481–500 msec, and ≥500 msec) will be summarized at baseline and at each subsequent time point. Categories of changes from baseline ≥30 and ≥60 msec will also be summarized.

All ECG findings include ECG abnormalities for each patient will be presented in a listing.

6.7.7 Ophthalmologic Examination

The following ophthalmologic examination will be reported for this study:

- Indirect Ophthalmoscopy Examination (Left/Right)
- Visual Acuity Test (Left/Right)
- Slit-Lamp Examination (Left/Right)

The ophthalmologic examination will be evaluated as normal/abnormal, not clinically significant (NCS)/abnormal, clinically significant (CS) by investigator.

The following summary table will be produced for ophthalmologic examination:

- Shift table from baseline to each post dose scheduled visit value according to abnormality

assessed by investigators (normal, CS and NCS).

Results of ophthalmologic examinations for each patient will be presented in a listing.

6.7.8 ECOG Performance Status

ECOG performance status will be presented using shift table to compare baseline to the worst post-baseline value.

ECOG performance status will also be presented for each patient in a data listing.

6.8 Pharmacokinetic Analysis

Blood samples for PK are to be collected as detailed protocol section 8.5.

Plasma concentrations of SNDX-5613 will be determined with a validated bioanalytical assay, the following parameters will be derived by conventional non-compartmental analysis (NCA) when sufficient data are available:

- $C_{\max}$ Maximum observed concentration
- $T_{\max}$ Time to maximum concentration
- AUC_{0-24} Area under the plasma concentration curve from time 0 to 24 hours
- AUC_{0-t} Area under the concentration versus time curve from time 0 to t
- CL/F Apparent oral clearance
- V_z/F Apparent volume of distribution
- $t_{1/2}$ Half-life

The concentration at 12 h after the morning dose (predose for evening dose) will be imputed with the concentration in the sample collected before the morning dose (predose for morning dose), as the values should be similar. AUC_{0-24} at steady state may be calculated as two times AUC_{0-12} with q12h dosing. Further details of the PK parameter calculations will be provided in a separate PK SAP.

Concentration-time data and PK parameters will be presented in listings of the individual values and will also be summarized by dose cohort/tumor type and day. Descriptive statistics to be reported for concentration-time data and estimated PK parameters include the number of non-missing observations (n), arithmetic mean, standard deviation, median, minimum, maximum, geometric mean, CV% of geometric mean (GeoCV). Concentration-time data and PK parameters of SNDX-5613 will be displayed to 3 significant figures in all data listings and summary tables, with the exception of time variable ($T_{\max}$) which will be displayed to 2 decimal places.

Coefficient of variation CV (%) is calculated as follows: $100 \times (\text{standard deviation}/\text{mean})$. Geometric CV (%) is calculated as follows: $CV (\%) = 100 \times \sqrt{\exp(\hat{\sigma}^2) - 1}$, where $\hat{\sigma}^2$ denotes the variance of the log-transformed values.

The PK population will be used for all summaries of the PK concentrations and parameters.

Individual plasma concentration will be plotted by dose cohort/tumor type, study day and visit in spaghetti plots.

The geometric mean of plasma concentration will also be plotted by dose cohort/tumor type, study day and visit.

6.9 Pharmacodynamics Analysis

Continuous summary (absolute and change from Baseline) of all the pharmacodynamics assessments will be presented by dose cohort/tumor type at all scheduled visits and time points per protocol.

Pharmacodynamics assessments will also be presented for each patient in a data listing. Actual sampling time points will be displayed for the assessments.

Based on data availability pharmacodynamic, and biomarker exploratory analyses will be described in a separate analysis plan and presented in a separate report.

6.10 Food Effect Analyses

Descriptive summaries of PK parameters will be provided for the fed and fasted state.

Natural log (ln) transformed PK parameters (AUC_{0-6} , and C_{max}) under fed and fasted conditions, and their difference, as well as their corresponding 90% confidence intervals (CIs) will be calculated. The geometric mean ratios (GMR) and their 90% CIs of the PK parameter will be determined. PK assessments for the fed and fasted state will also be presented for each patient in a data listing. Actual sampling time points will be displayed for the assessments.

6.11 Formulation Effect Analyses

Descriptive summaries of PK parameters will be provided for the tablet and capsule formulations.

Natural log (ln) transformed PK parameters (AUC_{0-12} , and C_{max}) under tablet and capsule conditions, and their difference, as well as their corresponding 90% confidence intervals (CIs) will be calculated. The geometric mean ratios (GMR) and their 90% CIs of the PK parameter will be determined. PK assessments for the tablet and capsule state will also be presented for each patient in a data listing. Actual sampling time points will be displayed for the assessments.

Details of PK evaluation analysis plan including formulation and food effect will be provided in a separate NCA analysis plan and provided as a separate technical report.

6.12 Interim Analysis

Toxicity monitoring is described in US Protocol Amendment 15.1 Version 14.1 Section 9.5.1.

An interim analysis for futility will be conducted for each cohort in phase 2. The investigator-assessed CR+CRh rate will be used as the endpoint for the interim analysis. The interim analysis

will be performed when the first 38 treated adult efficacy evaluable patients have had the opportunity to complete a minimum of 4 cycles of therapy or have discontinued therapy. Based on Simon's 2-stage design, if fewer than 5 responders are observed out of these initial 38 adult evaluable patients, patient enrollment will be discontinued in that cohort. Otherwise, patient enrollment will continue until a total of 64 adult efficacy evaluable patients are enrolled.

For Cohort 2B, a superiority/efficacy boundary will be added at the time of the interim analysis. If 10 or more adult AML efficacy evaluable patients achieve CR/CRh, the primary endpoint for Cohort 2B will be met and the study will be considered positive. If not, Cohort 2B will continue unless the futility boundary is met (ie, ≤ 4 adult AML efficacy evaluable patients achieve CR/CRh).

The hypothesis for the pooled population will also be tested at the time of the interim analysis for Cohort 2B. Approximately 58 efficacy evaluable patients are expected to be available from the pooled population. If 12 or more efficacy evaluable patients achieve CR/CRh, the hypothesis test for the pooled population is considered statistically significant. If the efficacy boundary is not met (ie, fewer than 12 CR/CRh responders), the hypothesis for the pooled population will be tested at the final analysis of Cohort 2B. The efficacy boundary will be re-calculated using the actual enrollment numbers at the time of the interim analysis, with alpha spending of approximately 0.01.

6.13 COVID-19 Related Analysis

The reasons for missing visits including relationship to COVID-19 if available may be listed.

7 CHANGES TO PLANNED ANALYSIS FROM STUDY PROTOCOL

- The start of DOR will be based on the investigator reported date of response in the primary analysis for DOR, not the date of bone marrow sampling.

8 REFERENCES

1. ICH Topic E3: Structure and Content of Clinical Study Reports (CPMP/ICH/137/95-adopted December 1995).
2. ICH Topic E9: Statistical Principles for Clinical Trials (CPMP/ICH/363/96 – adopted March 1998).
3. LeTourneau C, Lee JJ, Siu LL. Dose Escalation Methods in Phase I Cancer Clinical Trials.
4. Jung SH and Kim KM. (2004). On the estimation of the binomial probability in multistage clinical trials. *Statistics in Medicine* 23, 881-896.
5. Guidance for Industry: Acute Myeloid Leukemia: Developing Drugs and Biological Products for Treatment, <https://www.fda.gov/media/162362/download>

9 APPENDICES

Appendix 1 SAS Code for the Efficacy Analyses

SAS Code for the Efficacy Analyses			
Efficacy Endpoint	Statistical Analysis	SAS Procedures	SAS Code
Time to Event Endpoints			
- Event-Free Survival - Overall Survival - Duration of Response	Kaplan-Meier curves	LIFETEST	<pre>/* Kaplan-Meier */ proc lifetest data=adtte2 outsurv=outos timelist=(6,12,24) reduceout; time aval*cnsr(1); strata trtpn; run; Note: cnsr represents censoring variable (1=censored, 0=event) proc lifetest data=adtte2 notable method=KM conftype=loglog; time aval*cnsr(1); strata stratum1.. <stratum 3>/group= trt01pn; run; Note: cnsr represents censoring variable (1=censored, 0=event)</pre>
Categorical Endpoints			
- CR rate - CRc (CR+CRh+CRi+CRp) rate - ORR(CRc+PR+MLFS) rate	For one-way table, 95% exact Confidence intervals of each group using Clopper-Pearson method		<pre>/*exact CI for each group*/ proc freq data=adtte2; by trt01pn; tables Response/binomial(exact); run;</pre>